Stacked Tucker Decomposition With Multi-Nonlinear Products for Remote Sensing Imagery Inpainting

Shuang Xu, Jiangjun Peng, *Member, IEEE*, Teng-Yu Ji, Xiangyong Cao, Kai Sun, Rongrong Fei, and Deyu Meng

***Abstract*— In the field of remote sensing (RS) imaging, the occurrence of adverse meteorological conditions or sensor malfunctions can lead to missing data, posing a substantial impediment. Low-rank tensor decomposition has emerged as a promising strategy for resolving this issue, as it enables the integration of diverse data priors within a unified framework. Although various decomposition techniques, such as Tucker decomposition and tensor ring decomposition (TRD), have been developed based on multilinear products, they may not adequately capture the complex structure of RS imagery. Therefore, there is a need for tensor decompositions that incorporate nonlinear operations. To alleviate this challenge, a multi-nonlinear product is defined, which enables the construction of a nonlinear Tucker decomposition (NTD) model. To enhance the model's capability, a stacked Tucker decomposition (STD) model is formulated, by representing a tensor as the product of a core tensor and a collection of factor matrices along each mode, utilizing the multi-nonlinear product, which potentially regulates the distribution of singular values, thereby achieving a more accurate characterization of textures. The proposed model, integrated with total variation regularization, is subsequently applied to the task of RS imagery inpainting. Extensive experimental results demonstrate the superiority of the proposed model over state-of-the-art (SOTA) methods across various tasks. This validates its effectiveness and adaptability in mitigating the challenges associated with RS imagery inpainting. The code is available at https://github.com/shuangxu96/STDTV.**

***Index Terms*— Low-rank tensor completion, remote sensing (RS) imagery inpainting, Tucker decomposition.**

Received 27 April 2024; revised 22 August 2024; accepted 20 September 2024. Date of publication 26 September 2024; date of current version 9 October 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFA1004100; in part by China NSFC Projects under Grant 12201497, Grant 12201490, Grant 62272375, and Grant 12226004; in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2023A1515011358 and Grant 2024A1515011851; in part by Shaanxi Fundamental Science Research Project for Mathematics and Physics under Grant 22JSQ033 and Grant 23JSQ029; in part by China Postdoctoral Science Foundation Funded Project under Grant 2021M702621; in part by the Special Scientific Research Project of Shaanxi Provincial Department of Education under Grant 23JK0347; in part by the Natural Science Foundation of Shannxi Province, China, under Grant 2024JC-YBQN-0037; and in part by the Fundamental Research Funds for the Central Universities under Grant D5000240095. *(Corresponding authors: Jiangjun Peng; Xiangyong Cao.)*

Shuang Xu, Jiangjun Peng, and Teng-Yu Ji are with the School of Mathematics and Statistics, Northwestern Polytechnical University, Xi'an 710021, China, and also with the Research and Development Institute, Northwestern Polytechnical University in Shenzhen, Shenzhen 518063, China (e-mail: andrew.pengjj@gmail.com).

Xiangyong Cao is with the School of Electronic and Information Engineering and the Key Laboratory for Intelligent Networks and Network Security, Ministry of Education, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: caoxiangyong@mail.xjtu.edu.cn).

Kai Sun is with the School of Mathematics and Statistics, Xi'an Jiaotong University, Xi'an 710049, China.

Rongrong Fei is with the School of Electronic Information and Artificial Intelligence, Shaanxi University of Science and Technology, Xi'an 710021, China.

Deyu Meng is with the School of Mathematics and Statistics and the Key Laboratory of Intelligent Networks and Network Security, Ministry of Education, Xi'an Jiaotong University, Xi'an 710049, China, and also with Macao Institute of Systems Engineering, Macau University of Science and Technology, Macau, China.

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2024.3468718, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2024.3468718

I. Introduction

REMOTE sensing (RS) imagery is a crucial resource for Earth exploration and understanding, underpinning various applications, including target detection [1] and classification [2]. However, its utility is often limited by degraded quality, comprising noise [3], [4], data missing [5], [6] and low spatial resolution [7], [8]. Particularly, missing data can arise from adverse weather conditions, such as clouds that cover approximately 35% of the Earth's surface at any given moment [9], or from sensor malfunctions, including the defective detectors in band 6 of the aqua moderate resolution imaging spectroradiometer (MODIS) [10] or the permanently failed scan line corrector (SLC) of the landsat enhanced thematic mapper plus (ETM+) sensor [11]. These data gaps can significantly impact subsequent applications, necessitating the need to predict and reconstruct missing information. Consequently, the field of RS imagery inpainting has emerged to address this challenge, aiming to restore and complement missing data in optical RS images.

The problem can be typically solved by tensor decomposition, due to the low-rank property of RS imagery. Representative models include canonical polyadic decomposition (CPD), Tucker decomposition (TD), and tensor ring decomposition (TRD). The existing models, despite their distinct decomposition formats, rely on multilinear products. TD's multilinear product is known as the mode-n product or tensor-matrix product, as described in [12]. In contrast, TR's multilinear product is the tensor–tensor product. Currently, all known tensor decomposition models employ multilinear products for contracting factor matrices/tensors. It is widely

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

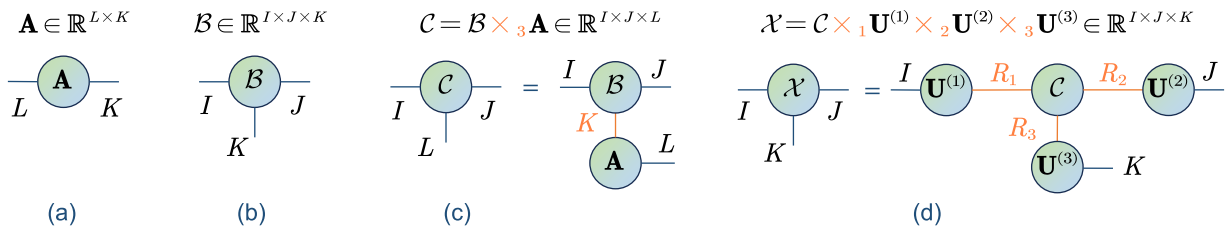


Fig. 1. Tensor diagrams for (a) matrix, (b) tensor, (c) multilinear product, and (d) Tucker decomposition.

recognized that multilinear products with more complex definitions can effectively leverage the inherent low-rank structure of observed data. This has led to the development of advanced factorization formats such as fully connected tensor networks (FCTNs) [5], [13], [14] and tensor wheel decomposition [15], which have demonstrated improved performance. However, these methods still primarily focus on exploring the linear relationships among different modes, potentially overlooking the nonlinearity of data structures.

The study presents a multi-nonlinear product, allowing for nonlinear contractions between the core tensor and factor matrices. A new Tucker decomposition model is formulated by replacing the conventional multilinear products with multi-nonlinear ones, aiming to enhance data reconstruction for complex data. To further enhance the model's nonlinearity, the study extends the vanilla Tucker decomposition to a stacked Tucker decomposition (STD) framework. In this model, the tensor is represented as a core tensor multiplied by a sequence of factor matrices for each mode, utilizing multi-nonlinear products. The numerical analyses exhibit that STD implicitly regulates the distribution of singular values with a pattern of few whopping singular values which represent coarse structures and a mass of mild singular values which stands for fine details. However, the increased nonlinearity in STD can lead to optimization challenges, such as local minima and overfitting. To mitigate these issues, a 3-D weighted total variation (3DWTV) regularization is incorporated into STD to ensure local smoothness in the reconstructed image. The main contributions of this work are as follows.

1) This work introduces the multi-nonlinear product as a novel operation for tensor-matrix contraction. Subsequently, a new tensor decomposition framework, termed STD, is developed by utilizing the multi-nonlinear product in the contraction of the core tensor with a sequence of factor matrices.
2) For RS imagery inpainting, the study presents the 3DWTV-regularized STD (STDTV) model, along with an optimization algorithm derived from gradient descent.
3) Comprehensive experiments demonstrate that the STDTV model outperforms state-of-the-art (SOTA) techniques in hyperspectral image (HSI) inpainting, multitemporal imagery (MTI) SCL-off inpainting, and MTI cloud removal tasks.

The following sections are organized as follows: Section II reviews related work on RS imagery inpainting. Section III introduces the necessary preliminaries. Section IV presents the development of the multi-nonlinear product and STDTV, along with the application to the RS imagery inpainting. Section V reports the experimental results, and Section VI provides a conclusion.

II. Related Work

RS imagery inpainting techniques can be classified into four categories: spatial, spectral, temporal, and hybrid methods.

The spatial-based methods operate akin to natural image inpainting, leveraging the interdependencies among local or nonlocal pixels in the spatial domain, without the need for spectral or temporal ancillary images. While traditional natural image inpainting techniques (including interpolation [16], total variation [17], propagated diffusion [18], and exemplar-based [19] methods) have been adapted for RS imagery inpainting to address missing data, their effectiveness in addressing extensive missing regions is limited by insufficient reference information. Nevertheless, spatial-based methods remain integral to RS imagery inpainting research, serving as a foundational stepping stone for the development of more advanced techniques.

Spectral-based methods for RS imagery inpainting aim to restore missing regions by leveraging supplementary information across different spectral bands. These methods estimate the interband relationships between intact and incomplete spectral areas, utilizing these established correlations for data interpolation. A notable example is the inpainting of aqua MODIS band 6, where numerous studies [10], [20], [21] have exploited a strong spectral correlation with band 7 to reconstruct the missing area. Similarly, other research [22], [23] has considered the relationships between band 6 and the remaining six bands. Spectral-based methods excel in reconstructing large areas and often outperform spatial-based methods. Nevertheless, their effectiveness can be limited when all bands of an RS image exhibit identical missing regions, as spectral-based methods may then struggle to produce satisfactory outcomes.

Temporal-based methods operate on MTI, using sequences of images captured over varying time intervals. It assumes a stable relationship between the missing area and the observed areas across time nodes. Temporal-based methods are typically used in cloud removal, given the varying cloud cover across different periods, which results in distinct missing regions within the images. They often involve filling the missing area with corresponding regions from different time intervals [9]. Recent advancements in matrix completion have leveraged low-rank techniques to recover missing information [24], [25], [26], [27].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

Hybrid methods, which integrate spatial, spectral, and temporal information concurrently, have recently emerged as a novel technique in RS imagery inpainting. By exploiting multiple types of prior information, these methods have the potential to overcome the limitations of traditional techniques. Low-rank tensor completion is an effective framework for integrating all priors and has been applied to this problem. The smooth PARAFAC tensor completion with total variation regularization (SPCTV) developed by Yokota et al. [28] is an early work that uses CPD to recover missing entries in high-dimensional visual data. CPD can be considered a special case of TD [12]. Ji et al. [6] and Ng et al. [29] used TD to explore spatial, spectral, and temporal information by unfolding the tensor into matrices along various dimensions and summing the ranks of the entire matrices. However, Tucker rank-based methods have drawbacks, such as an unbalanced matricization scheme that may not fully capture the global information of the tensor. Zhang et al. [30] explored tensor train decomposition (TTD)-based RS imagery inpainting, trying to solve the problem that traditional TTD uses the tensor augmentation trick for modeling spatial correlation. They proposed a full-mode-augmentation strategy and achieved modeling spatial–spectral interaction. He et al. [31] introduced the application of TRD [32], [33], [34], [35] with TV regularization, aiming to overcome the limitations of TD. TRD represents the tensor through circular multilinear products over a sequence of third-order core tensors, enabling the method to model relationships between adjacent modes. More recently, Zheng et al. [5] proposed a novel FCTN framework for multidimensional image processing, and it has been applied to HSI inpainting by the nonlocal variant [13], [36] and MTI cloud removal by the spatial-spectral–temporal extension [14].

III. Preliminaries

A. Notations and Diagrams

Matrices are represented by boldface capital letters, for example, $\mathbf{A} \in \mathbb{R}^{L\times K}$, and they are visualized as nodes with two lines annotated with the dimensions of each mode, as exemplified in Fig. 1(a). Tensors are denoted by Euler script letters, for instance, $\mathcal{B} \in \mathbb{R}^{I\times J\times K}$, and they are represented as nodes with three lines annotated with the dimensions of each mode, as displayed in Fig. 1(b). The tensor-matrix multilinear product, also known as the mode-n product, is denoted as $\mathcal{C} = \mathcal{B} \times_n \mathbf{A}$. For instance, the mode-3 product results in $\mathcal{C} \in \mathbb{R}^{I\times J\times L}$, and the (i, j, l) entry is defined by the equation

$$c_{ijl} = \sum_{k=1}^{K} b_{ijk} a_{lk}. \tag{1}$$

In tensor diagrams, this operation is illustrated using an orange line connecting a matrix and a tensor, as shown in Fig. 1(c).

TD represents a tensor $\mathcal{X} \in \mathbb{R}^{I\times J\times K}$ as

$$\begin{aligned}\mathcal{X} &= [\![\mathcal{C}; \mathbf{U}^{(1)}, \mathbf{U}^{(2)}, \mathbf{U}^{(3)}]\!] \\ &= \mathcal{C} \times_1 \mathbf{U}^{(1)} \times_2 \mathbf{U}^{(2)} \times_3 \mathbf{U}^{(3)}\end{aligned} \tag{2}$$

with

$$x_{ijk} = \sum_{r_1=1}^{R_1}\sum_{r_2=1}^{R_2}\sum_{r_3=1}^{R_3} c_{r_1r_2r_3} u^{(1)}_{ir_1} u^{(2)}_{jr_2} u^{(3)}_{kr_3} \tag{3}$$

where $\mathcal{C} \in \mathbb{R}^{R_1\times R_2\times R_3}$ is the core tensor, and $\mathbf{U}^{(1)} \in \mathbb{R}^{I\times R_1}$, $\mathbf{U}^{(2)} \in \mathbb{R}^{J\times R_2}$, and $\mathbf{U}^{(3)} \in \mathbb{R}^{K\times R_3}$ are the factor matrices. TD is visualized using a tensor network diagram in Fig. 1(d), where each mode of the core tensor is connected to its corresponding factor matrix via an orange line.

B. RS Imagery Inpainting

Due to nonideal weather conditions (e.g., clouds) and hardware limitations (e.g., SLC off), RS images are frequently observed only partially. The goal of inpainting is to reconstruct the unobserved values using the observed ones as guidance. Given a partially observed RS image $\mathcal{Y} \in \mathbb{R}^{H\times W\times C}$, the binary tensor $\mathcal{W} \in \{0, 1\}^{H\times W\times C}$ is employed to indicate whether each entry is observed, where $w_{ijk} = 1$ signifies that the (i, j, k)th entry is observed, and 0 otherwise. The index set Ω comprises the indices of observed entries: $\Omega = \{(i, j, k) \mid w_{ijk} = 1\}$. The orthogonal projection operator is defined as

$$[\mathcal{P}_\Omega(\mathcal{X})]_{ijk} = \begin{cases} x_{ijk}, & (i, j, k) \in \Omega \\ 0, & (i, j, k) \notin \Omega. \end{cases} \tag{4}$$

In other words, the operator returns zero for unobserved (i, j, k) entries and the original value otherwise. Typically, RS imagery inpainting can be addressed by solving the following optimization problem:

$$\min_{\mathcal{X}} \psi_{\mathbb{S}_\mathcal{Y}}(\mathcal{X}) + R(\mathcal{X}) \tag{5}$$

where $R(\mathcal{X})$ represents the regularization term applied to $\mathcal{X}$. The first term is defined by

$$\psi_{\mathbb{S}_\mathcal{Y}}(\mathcal{X}) = \begin{cases} 0, & \text{if } \mathcal{X} \in \mathbb{S} \\ \infty, & \text{otherwise} \end{cases} \tag{6}$$

with $\mathbb{S}_\mathcal{Y} = \{\mathcal{X} \mid \mathcal{P}_\Omega(\mathcal{X} - \mathcal{Y}) = 0\}$, ensuring that the reconstructed image $\mathcal{X}$ maintains the same values as the partially observed image $\mathcal{Y}$ for observed pixels.

IV. Multi-Nonlinear Product and STD

A. Multi-Nonlinear Product

This study aims to establish a novel nonlinear tensor decomposition model. The fundamental operation within tensor decomposition involves the contraction of factor tensors or matrices, namely mode-n products (tensor-matrix products) or tensor–tensor products. To the best of our knowledge, existing methods are constructed using multilinear products. Consequently, we introduce the multi-nonlinear product, which can be seen as a nonlinear extension of the mode-n product.

Definition 1: Let $\mathcal{C} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$ be a tensor of order N, $\mathbf{U} \in \mathbb{R}^{J\times I_n}$ a matrix, and $\sigma(\cdot)$ a nonlinear function, the (tensor-matrix) multi-nonlinear product is defined as

$$\mathcal{X} = \sigma(\mathcal{C} \times_n \mathbf{U}) \tag{7}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

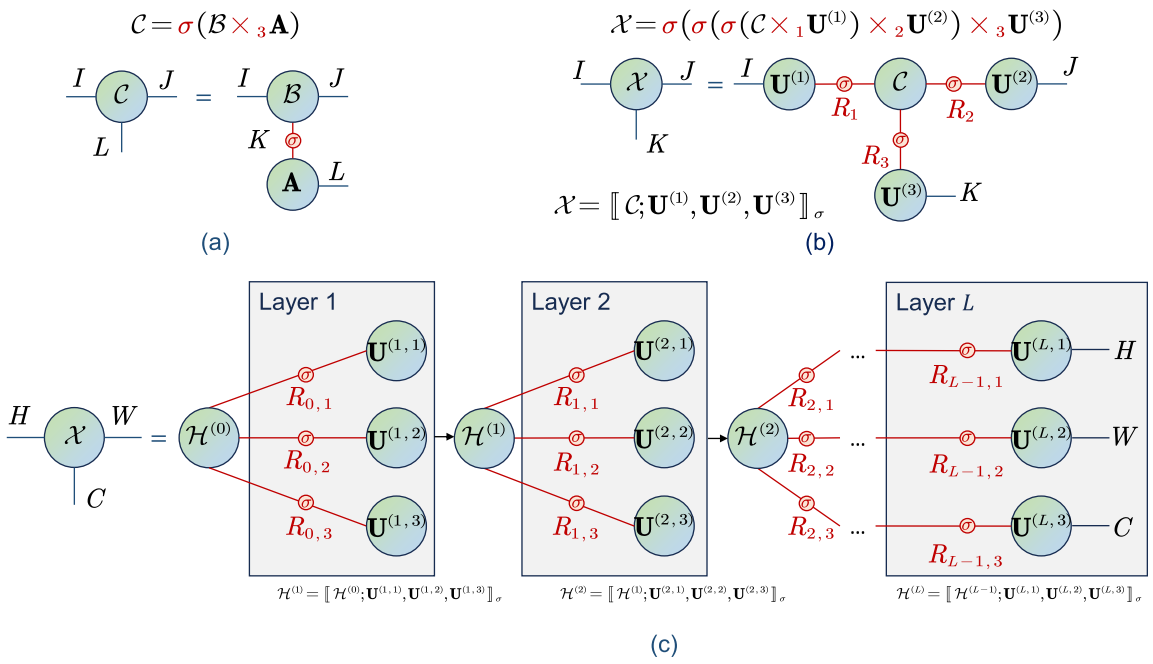


Fig. 2. Tensor diagrams for (a) multi-nonlinear product, (b) tucker decomposition with multi-nonlinear product (i.e., NTD), and (c) STD.

where $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_N}$, and its element is given by

$$x_{i_1,\ldots,i_{n-1}ji_{n+1},\ldots,i_N} = \sigma\left(\sum_{i_n=1}^{I_n} c_{i_1 i_2 \cdots i_N} u_{ji_n}\right). \tag{8}$$

Based on this definition, various multi-nonlinear products can be derived by selecting different functions $\sigma(\cdot)$. If $\sigma(\cdot)$ is the identity function, the multi-nonlinear product degrades to the multilinear product. In tensor diagrams, the multi-nonlinear product is represented by a red line connecting a matrix and a tensor, with a circled σ symbol on this line, as illustrated in Fig. 2(a).

In the following experiments, $\sigma(\cdot)$ is configured by Mish [37], a self-regularized nonmonotonic activation function which has shown better performance than classic activation functions (e.g., the rectified linear unit and its variants), defined as

$$\text{Mish}(x) = x\text{Tanh}(\text{Softplus}(x)) \tag{9}$$

with $\text{Softplus}(x) = \ln(1 + e^x)$, and $\text{Tanh}(\cdot)$ being a hyperbolic tangent function.

B. Stacked Tucker Decomposition

The Tucker decomposition with the multi-nonlinear product is ready for construction and is hereinafter referred to as the nonlinear Tucker decomposition (NTD). Specifically, consider a third-order tensor $\mathcal{X}$ with multirank $[r_1, r_2, r_3]$, and its NTD model is represented as follows:

$$\begin{aligned}\mathcal{X} &= [\![\mathcal{C}; \mathbf{U}^{(1)}, \mathbf{U}^{(2)}, \mathbf{U}^{(3)}]\!]_\sigma \\ &= \sigma\left(\sigma\left(\sigma\left(\mathcal{C} \times_1 \mathbf{U}^{(1)}\right) \times_2 \mathbf{U}^{(2)}\right) \times_3 \mathbf{U}^{(3)}\right).\end{aligned} \tag{10}$$

Equivalently, the element of $\mathcal{X}$ is expressed by

$$x_{ijk} = \sigma\left(\sum_{r_3=1}^{R_3} \sigma\left(\sum_{r_2=1}^{R_2} \sigma\left(\sum_{r_1=1}^{R_1} c_{r_1 r_2 r_3} u_{ir_1}^{(1)}\right) u_{jr_2}^{(2)}\right) u_{kr_3}^{(3)}\right). \tag{11}$$

The NTD represents a tensor as the core tensor $\mathcal{C}$ multiplied by a group of factor matrices along each mode using the multi-nonlinear product. As illustrated in Fig. 2(b), the nonlinear version of the Tucker decomposition replaces the multilinear product with the multi-nonlinear product.

Remark that both the vanilla Tucker decomposition and its nonlinear variant primarily capture the superficial characteristics of tensor data. However, does this straightforward framework adequately delineate the rich contents for RS imagery? Studies in neuroscience have established that the human visual system employs a multilayered, nonlinear processing mechanism to interpret images [38]. Consequently, the models described by (2) and (10) do not align with the operational principles of the human visual system.

To solve this issue, a STD is formulated in a multilayered fashion, and the L-layer STD is expressed by

$$\begin{aligned}\mathcal{H}^{(1)} &= [\![\mathcal{H}^{(0)}; \mathbf{U}^{(1,1)}, \mathbf{U}^{(1,2)}, \mathbf{U}^{(1,3)}]\!]_\sigma \\ \mathcal{H}^{(2)} &= [\![\mathcal{H}^{(1)}; \mathbf{U}^{(2,1)}, \mathbf{U}^{(2,2)}, \mathbf{U}^{(2,3)}]\!]_\sigma \\ &\cdots \\ \mathcal{H}^{(L)} &= \mathcal{X} = [\![\mathcal{H}^{(L-1)}; \mathbf{U}^{(L,1)}, \mathbf{U}^{(L,2)}, \mathbf{U}^{(L,3)}]\!]_\sigma\end{aligned} \tag{12}$$

where $\mathcal{H}^{(0)} \in \mathbb{R}^{R_{0,1} \times R_{0,2} \times R_{0,3}}$ denotes the core tensor, $\mathcal{H}^{(l)} \in \mathbb{R}^{R_{l,1} \times R_{l,2} \times R_{l,3}} (l = 1, 2, \ldots, L-1)$ is the latent tensor at each layer, serving as an intermediate output, $\mathcal{X} = \mathcal{H}^{(L)} \in \mathbb{R}^{H \times W \times C}$ is the final output tensor, and $\mathbf{U}^{(l,n)} \in \mathbb{R}^{R_{l,n} \times R_{l-1,n}}$ denotes the factor matrix of the lth layer along the nth mode. The STD defined in (12) is succinctly abbreviated as

$$\mathcal{X} = \text{STD}\left(\mathcal{H}^{(0)}, \left\{\mathbf{U}^{(l,n)}\right\}_{l=1,n=1}^{L,3}\right). \tag{13}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

The STD exhibits compatibility with the human visual system's processing mechanisms [38], [39]. Furthermore, STD's nonlinear nature endows it with enhanced data fitting capabilities, making it particularly well-suited for processing RS imagery characterized by complex and rich information content.

C. Application to RS Imagery Inpainting

The STD is subsequently employed for RS imagery inpainting. Due to STD's complexity and nonlinearity, optimization may converge to local minima, yielding suboptimal results. Thus, we integrate a 3DWTV regularizer to enhance local smoothness. Consequently, the 3DWTV-regularized STD inpainting problem is posed as

$$\min_{\mathcal{X},\mathcal{H}^{(0)},\mathbf{U}^{(l,m)}} f(\mathcal{X},\mathcal{H}^{(0)},\mathbf{U}^{(l,m)}) = \left\{\|\mathcal{X}-\mathcal{H}^{(L)}\|_F^2+\alpha\|\mathcal{H}^{(L)}\|_{3\text{DWTV}}+\psi_{\mathbb{S}_\mathcal{Y}}(\mathcal{X})\right\} \quad (14)$$

with

$$\mathcal{H}^{(L)} = \text{STD}\left(\mathcal{H}^{(0)},\left\{\mathbf{U}^{(l,m)}\right\}_{l=1,n=1}^{L,3}\right)$$

and $\|\mathcal{H}^{(L)}\|_{3\text{DWTV}} = \sum_{i=1}^{3}\beta_i\|\nabla_i\mathcal{H}^{(L)}\|_1$, where ∇_i is the gradient operator along mode-i, $\psi_{\mathbb{S}_\mathcal{Y}}(\mathcal{X})$ is defined in (6), and α and β_i are the nonnegative weights. Here, β_1 and β_2 control the spatial smoothness, while β_3 controls the spectral one, so generally let $\beta_1=\beta_2=1$ and β_3 is tuning.

The unknown variables $\mathcal{X}$, $\mathcal{H}^{(0)}$, and $\mathbf{U}^{(l,m)}$ are iteratively updated, and their respective updating rules are defined as follows:

$$\mathcal{H}^{(0)} \leftarrow \mathcal{H}^{(0)} - \eta\partial f/\partial\mathcal{H}^{(0)} \quad (15)$$

$$\mathbf{U}^{(l,m)} \leftarrow \mathcal{H}^{(0)} - \eta\partial f/\partial\mathbf{U}^{(l,m)} \quad (16)$$

$$\mathcal{H}^{(L)} \leftarrow \text{STD}\left(\mathcal{H}^{(0)},\left\{\mathbf{U}^{(l,m)}\right\}_{l=1,n=1}^{L,3}\right) \quad (17)$$

$$\mathcal{X} \leftarrow \mathcal{P}_\Omega(\mathcal{Y})+\mathcal{P}_{\Omega^c}(\mathcal{H}^{(L)}) \quad (18)$$

where η represents the learning rate, and Ω^c is the complement of Ω. In these equations, $\mathcal{H}^{(0)}$ and $\mathbf{U}^{(l,m)}$ are updated utilizing the gradient descent algorithm.

D. Rank Configuration

In STD, $R_{l,n}(l=0,1,\ldots,L-1;n=1,2,3)$ is named STD-rank, parameterized by a $L\times 3$ matrix. The ranks of the last layer $(R_{L-1,1},R_{L-1,2},R_{L-1,3})$ actually equals to the size of the decomposed tensor, (H,W,C). Therefore, $3(L-1)$ parameters to be configured, and it will lead to laborious parameter selection. After extensive experiments, we developed a strategy to alleviate this issue. The rank of the first layer, $(R_{0,1},R_{0,2},R_{0,3})$, is first set. Subsequently, other ranks is configured by uniformly sampling from the interval $[R_{0,1},R_{L-1,1}]$. For example, given a tensor with a size of $100\times 100\times 30$ and a three-layer STD model, it first set $(R_{0,1},R_{0,2},R_{0,3})=(20,20,10)$. Then, according to the proposed strategy, $(R_{1,1},R_{1,2},R_{1,3})$ should be $(60,60,20)$. With strategy, we only need to determine three parameters $(R_{0,1},R_{0,2},R_{0,3})$, instead of $3(L-1)$ parameters.

V. Experiments

This study mainly focuses on three kinds of RS imagery inpainting tasks, comprising HSI random inpainting, MTI deadline inpainting, and MTI cloud removal. We employ peak signal-to-noise ratio (PSNR) and structural similarity (SSIM), erreur relative global adimensionnelle de synthèse (ERGAS) and spectral angle mapper (SAM) as the evaluation metrics for all experiments. SOTA methods are selected for comparison, comprising, the sum of the nuclear norm (SNN)[1] [40], TV regularized SNN (SNNTV) [41], SPCTV[2] [28], tensor nuclear norm (TNN)[3] [42], fully connected tensor network (FCTN)[4] [5], tensor correlated TV (tCTV)[5] [43], S2NTNN[6] [44], and tensor-Q rank (TQN) [45].

A. Synthetic Experiments

1) HSI Random Inpainting: The Xiongan dataset[7] [46] was acquired by the visible and near-infrared imaging spectrometer designed by Shanghai Institute of Technical Physics, Chinese Academy of Sciences in October 2017. It contains 250 bands with a spectral range from 400 to 1000 nm. For simulations, a patch of 256×256 pixels was cropped. The experiments set very high missing rates, including 98%, 99%, and 99.5%.

The STD-rank of the first layer, denoted as $(R_{0,1},R_{0,2},R_{0,3})$, was established as $(0.6H,0.6W,0.1B)$ where H, W, and B represent the height, width, and band number of the Xiongan dataset, respectively. The STD architecture comprised of two layers. The trade-off parameter for 3DWTV, denoted as α, was set to 10^{-4}, while the parameter vector $\boldsymbol{\beta}$ was assigned values of $(1,1,1)$. Additionally, the learning rate was fixed at 10^{-3}.

Fig. 3 presents the composited images with bands 119-56-31 of the Xiongan dataset before and after reconstruction in three scenarios. Visual inspection reveals that SNN struggled to recover structural details, while SNNTV and TNNTV struggled with fine-coarse texture reconstruction. SPCTV and TNN provided visually appealing results, except in the 99% and 99.5% missing rate cases. Although tCTV performed second best, it omitted some crucial details. In conclusion, STDTV offers the most favorable visual outcomes.

Table I presents the quantitative evaluation results of all completion methods for HSI random missing data. The best results are highlighted in bold. According to the table, the proposed STDTV achieved the most favorable outcomes across all scenarios. At a missing rate of 98%, most methods achieved a PSNR value below 33 dB, except for tCTV (36.37 dB), TQN (37.4 dB), and STDTV (40.77 dB), which retained impressive performance. However, the PSNR values of tCTV and TQN decreased to 32.71 and 30.36 dB for the 99% missing rates. In contrast, STDTV maintained good performance with a PSNR of 37.65 dB.

[1] https://www.cs.rochester.edu/u/jliu/publications.html
[2] https://ieeexplore.ieee.org/document/7502115/media#media
[3] https://github.com/canyilu/tensor-completion-tensor-recovery
[4] https://yubangzheng.github.io/
[5] https://github.com/wanghailin97/Guaranteed-Tensor-Recovery-Fused-Low-rankness-and-Smoothness
[6] https://github.com/YisiLuo/S2NTNN
[7] http://www.hrs-cas.com/a/share/shujuchanpin/2019/0501/1049.html

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE EVALUATION OF SIMULATED DATA EXPERIMENTS FOR HSI RANDOM INPAINTING WITH MISSING RATES (MR) OF 98%, 99%, AND 99.5%. THE BEST, SECOND BEST, AND THIRD BEST RESULTS ARE MARKED BY **BOLD**, UNDERLINE AND ITALIC, RESPECTIVELY

MR	Metrics	Observed	SNN	SNNTV	SPCTV	TNN	FCTN	tCTV	S2NTNN	TQN	STDTV
98%	PSNR↑	8.47	23.09	25.64	28.02	27.47	32.66	*36.37*	27.53	37.4	**40.77**
	SSIM↑	0.0071	0.5904	0.6075	0.7236	0.6989	0.8298	*0.9307*	0.7626	0.9428	**0.9689**
	ERGAS↓	5042	22.29	16.17	11.87	11.65	6.81	*4.29*	11.73	3.96	**2.5**
	SAM↓	82.17	5.12	3.83	3.4	4.59	2.26	*1.65*	3.11	1.39	**0.92**
99%	PSNR↑	8.43	20.83	21.51	24.67	18.11	29.01	32.71	23.84	*30.36*	**37.65**
	SSIM↑	0.0043	0.569	0.576	0.613	0.3273	0.6828	0.8597	*0.8306*	0.791	**0.9403**
	ERGAS↓	10150	29.43	26.42	17.78	49.58	10.2	6.61	18.28	*8.98*	**3.68**
	SAM↓	84.8	5.72	5.99	4.77	11.38	3.65	2.29	5.44	*2.95*	**1.2**
99.5%	PSNR↑	8.41	18.94	9.13	23.24	12.09	22.32	29.75	*25.7*	24.76	**34.09**
	SSIM↑	0.0026	0.5537	0.0927	0.5871	0.1812	0.3267	0.7684	*0.6399*	0.5697	**0.8782**
	ERGAS↓	20428.32	39.83	1085.18	21.36	193.19	23.25	9.2	*14.92*	17.08	**5.69**
	SAM↓	86.78	7.12	30.83	5.38	20.71	9.9	3.02	*3.63*	5.52	**1.73**

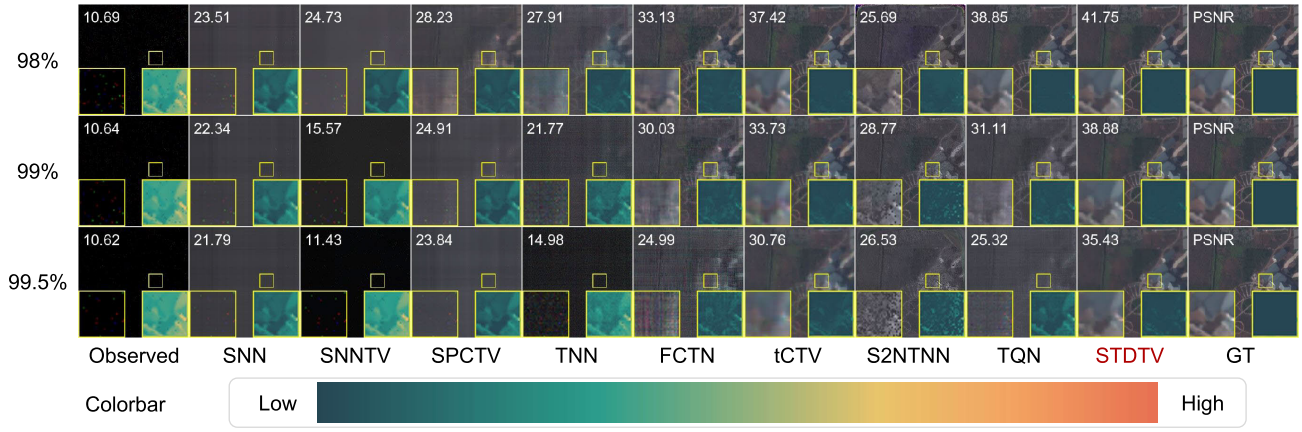


Fig. 3. HSI restoration results of different methods (bands: 119-56-31) on Xiongan dataset. (From top to bottom) There are three cases with different missing rates. (Top left) PSNR value of these bands. (Bottom left) and (bottom right) Amplified area and its error map.

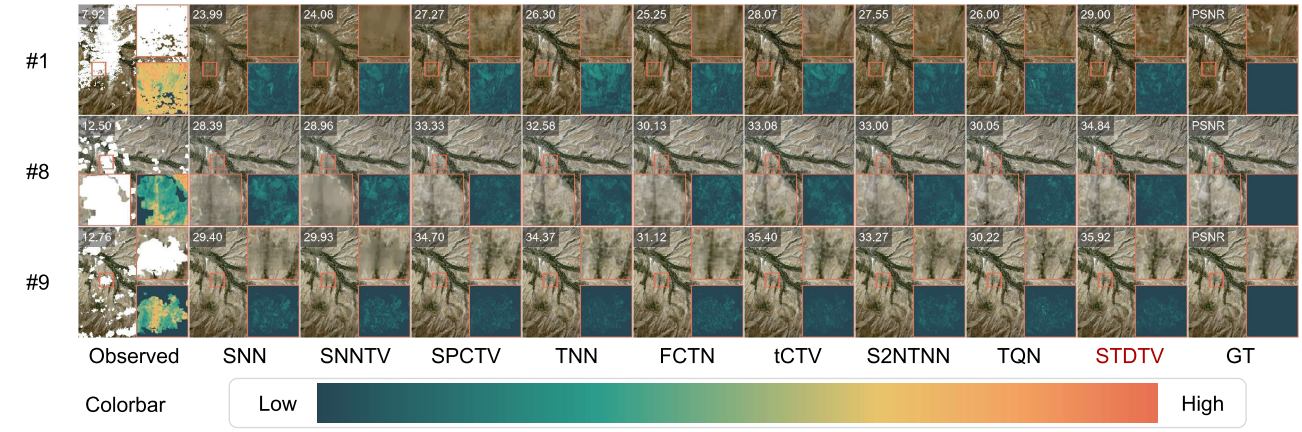


Fig. 4. MTI cloud removal results of different methods (bands: 4-3-2) on the Jizzakh dataset. (From top to bottom) There are three time-nodes. (Top left) PSNR value of these bands. (Top-right) and (bottom right) Amplified area and its error map.

2) MTI Cloud Removal: Fig. S1 presents the Jizzakh dataset, acquired by the Landsat-8 satellite over Jizzakh, Uzbekistan. The dataset consists of 512 × 512 pixels and seven spectral bands (coastal, blue, green, red, near-infrared, shortwave infrared 1, and shortwave infrared 2). It covers nine different temporal instances: 2013-07-15, 2013-08-16, 2013-09-01, 2013-09-17, 2013-10-03, 2013-10-19, 2014-05-15, 2014-07-18, and 2014-08-19. This results in a total data size of 512 × 512 × 7 × 9. To simulate the presence of thick clouds, nine distinct cloud masks were generated with varying levels of cloud coverage.

The STD-rank of the first layer, denoted as $(R_{0,1}, R_{0,2}, R_{0,3})$, was established with parameters $(0.4H, 0.4W, 0.3B)$, where H, W, and B represent the height, width, and band count of the Jizzakh dataset, respectively. The STD architecture comprised four layers.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE EVALUATION OF SIMULATED DATA EXPERIMENTS FOR MTI CLOUD REMOVAL FOR NINE TIME NODES. THE BEST, SECOND BEST, AND THIRD BEST RESULTS ARE MARKED BY **BOLD**, UNDERLINE, AND ITALIC, RESPECTIVELY

Time node	Metrics	Observed	SNN	SNNTV	SPCTV	TNN	FCTN	tCTV	S2NTNN	TQN	STDTV
#1	PSNR↑	10.96	24.47	24.95	27.72	27.07	25.58	28.29	*28.09*	26.43	**29.61**
	SSIM↑	0.5059	0.7853	0.7726	0.8766	0.8951	0.7924	**0.9135**	*0.9001*	0.861	0.913
	ERGAS↓	117.54	18.82	18.83	13.47	14	16.99	*12.27*	12.13	14.66	**10.46**
	SAM↓	35.27	2.68	2.48	*2.01*	2.01	2.57	1.79	2.06	2.57	**1.56**
#2	PSNR↑	10.25	25.35	26.01	30.35	30.31	26.68	*31.7*	**32.47**	28.7	32.46
	SSIM↑	0.5313	0.8382	0.8328	0.9302	0.9378	0.8457	*0.9525*	**0.9583**	0.9138	0.957
	ERGAS↓	86.01	11.55	11.07	6.78	6.85	10.04	*5.91*	**5.34**	8.35	5.4
	SAM↓	29.16	1.71	1.51	*1.13*	1.19	1.63	1.09	1.19	1.89	**0.92**
#3	PSNR↑	11.37	28.03	29.04	34.38	33.84	29.58	35.62	*35.53*	31.32	**36.96**
	SSIM↑	0.6543	0.9077	0.9074	0.967	0.9669	0.9063	*0.9762*	0.9785	0.946	**0.9809**
	ERGAS↓	61.79	7.72	7.05	3.69	3.98	6.39	3.23	*3.34*	5.56	**2.79**
	SAM↓	19.39	0.95	0.8	*0.61*	0.63	0.92	0.55	0.88	1.46	**0.53**
#4	PSNR↑	12.42	26.99	27.95	35.14	33.32	29.83	*35.31*	37.35	34.29	**38.59**
	SSIM↑	0.7171	0.8936	0.8896	0.9716	0.965	0.9087	*0.9759*	0.9835	0.9688	**0.9854**
	ERGAS↓	55.4	8.75	7.98	3.42	4.32	6.25	*3.4*	2.79	3.96	**2.33**
	SAM↓	20.25	1.2	0.95	*0.71*	0.74	0.98	0.66	0.88	1.29	**0.55**
#5	PSNR↑	8.38	21.42	22.15	30.49	28.78	25.41	30.67	34.55	*33.56*	**34.95**
	SSIM↑	0.477	0.7565	0.7338	0.9413	0.9327	0.8121	0.9524	0.971	*0.9687*	**0.9745**
	ERGAS↓	118.87	16.31	15.26	5.56	7.02	10.02	5.51	3.52	*4.1*	**3.35**
	SAM↓	39.33	2.5	2.15	1.3	1.57	1.85	1.39	1.02	*1.28*	**0.9**
#6	PSNR↑	9.12	21.18	21.78	29.08	25.95	25.13	26.67	32.53	*30.16*	**33.65**
	SSIM↑	0.6147	0.814	0.7946	0.9463	0.9175	0.8614	0.9258	0.9705	*0.9551*	**0.9756**
	ERGAS↓	92.82	16.41	15.36	6.33	9.12	10.08	8.39	4.41	*5.7*	**3.73**
	SAM↓	28.66	1.71	1.47	1.01	1.08	1.37	*1.04*	1.14	1.46	**0.77**
#7	PSNR↑	17.03	32.96	33.87	*36.41*	35.14	34.28	35.39	36.44	33.97	**37.15**
	SSIM↑	0.9094	0.9708	0.9697	0.9813	0.9814	0.9701	0.9832	*0.983*	0.9698	**0.9835**
	ERGAS↓	32.21	5.68	5.23	*3.73*	4.39	4.83	4.3	3.61	4.55	**3.39**
	SAM↓	5.56	0.36	0.33	**0.3**	0.31	0.35	*0.31*	0.82	1.2	0.4
#8	PSNR↑	12.6	28.64	29.54	*33.28*	32.77	30.07	33.27	33.59	30.67	**35.03**
	SSIM↑	0.7221	0.9092	0.9078	0.9535	0.9561	0.911	*0.9623*	0.9658	0.9354	**0.9676**
	ERGAS↓	55.1	7.63	7.13	*4.56*	4.95	6.5	4.78	4.55	6.23	**3.76**
	SAM↓	17.08	0.94	0.83	0.61	0.68	0.87	*0.66*	1.02	1.48	**0.6**
#9	PSNR↑	13.52	29.35	30.33	34.57	33.89	30.98	*34.46*	33.75	30.92	**36.12**
	SSIM↑	0.7474	0.9286	0.9277	0.9674	0.9674	0.928	0.9726	*0.9719*	0.9407	**0.9774**
	ERGAS↓	49.88	7.47	6.98	*4.26*	4.3	6.22	4	4.82	6.49	**3.52**
	SAM↓	14.89	0.89	0.78	0.65	*0.61*	0.83	0.58	1.06	1.51	**0.57**
Mean	PSNR↑	11.74	26.49	27.29	*32.38*	31.23	28.61	32.37	33.81	31.12	**34.95**
	SSIM↑	0.6532	0.8671	0.8595	0.9484	0.9467	0.8817	*0.9572*	0.9647	0.9399	**0.9683**
	ERGAS↓	74.4	11.15	10.54	5.76	6.55	8.59	*5.75*	4.95	6.62	**4.3**
	SAM↓	23.29	1.44	1.26	*0.92*	0.98	1.26	0.9	1.12	1.57	**0.76**

The trade-off parameter for the 3DWTV term were set at $\alpha = 10^{-3}$, with $\boldsymbol{\beta} = (1, 1, 0.05)$, and the learning rate was fixed at 10^{-3}.

Fig. 4 displays the reconstruction outcomes at the first, eighth, and ninth time nodes for comparison. It is evident that SNNTV and TNNTV exhibit excessive smoothing, while SNN and TNN are capable of recovering coarse structures. Although SPCTV and tCTV retrieve missing information, they fail to preserve textures.

Table II presents the quantitative assessment of all completion techniques for the cloud removal problem. Notably, the most favorable results are highlighted in bold. The proposed STDTV achieved a PSNR value of 34.95 dB, and yielded the superior performance across all methods, with an improvement of 1.14 dB compared with S2NTNN.

3) MTI SLC-Off Inpainting: The dataset for MTI deadline inpainting, termed Puyang dataset, was acquired by the Landsat-8 satellite from Puyang, Henan, China. It comprises 512×512 pixels, seven bands (coastal, blue, green, red, near-infrared, shortwave infrared 1, and shortwave infrared 2), and nine time nodes (2013-04-26, 2013-05-12, 2015-05-18, 2015-09-07, 2015-10-09, 2016-04-18, 2016-05-04, 2016-11-28, and 2017-09-28), leading to a size of $512 \times 512 \times 7 \times 9$. To simulate the SLC-off phenomenon, several wide deadlines were utilized to introduce corruption into this dataset. Fig. S2 provides visualizations of the MTIs and their corresponding masks.

The STD-rank of the first layer, denoted as $(R_{0,1}, R_{0,2}, R_{0,3})$, was set as $(0.2H, 0.2W, 0.1B)$, where H, W, and B represent the height, width, and band number of the Puyang dataset, respectively. The STD model consisted of five layers. The trade-off parameter of the 3DWTV regularization term α was set to 10^{-2}, weight $\boldsymbol{\beta}$ was $(1, 1, 0.05)$, and the learning rate was 10^{-3}.

Fig. 5 illustrates the RGB band representation of the Puyang dataset at three distinct time nodes. It clearly indicates that the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE EVALUATION OF SIMULATED DATA EXPERIMENTS FOR MTI SLC-OFF INPAINTING FOR NINE TIME NODES

Time node	Metrics	Observed	SNN	SNNTV	SPCTV	TNN	FCTN	tCTV	S2NTNN	TQN	STDTV
#1	PSNR↑	11.08	20.79	21.02	*25.62*	21.39	22.39	22.33	27.13	23.24	**28.54**
	SSIM↑	0.5412	0.7507	0.7451	*0.8798*	0.7825	0.7852	0.8096	0.9218	0.8464	**0.9331**
	ERGAS↓	110.97	25.73	25.46	*14*	23.47	20.76	21.35	12.11	18.6	**9.98**
	SAM↓	32.58	4.72	4.49	*2.7*	5.12	3.75	4.71	2.51	4.38	**1.99**
#2	PSNR↑	10.4	20.44	20.64	*25.12*	22.27	21.95	23.59	26.72	23.25	**27.94**
	SSIM↑	0.523	0.7386	0.7319	*0.8713*	0.8123	0.7715	0.8507	0.9232	0.8425	**0.9276**
	ERGAS↓	117.66	26.57	26.5	*14.56*	19.88	21.55	16.77	12.53	18.06	**10.5**
	SAM↓	33.73	4.63	4.43	*2.78*	4.14	3.82	3.52	2.52	4.5	**2.06**
#3	PSNR↑	10.28	20.17	20.29	*22.12*	21.19	21.27	21.97	22.19	19.84	**23.12**
	SSIM↑	0.5254	0.7417	0.732	*0.8203*	0.787	0.7702	0.8188	0.822	0.752	**0.8403**
	ERGAS↓	115.21	25.76	25.91	19.58	22.13	21.96	*20.14*	20.26	25.3	**18.37**
	SAM↓	33.54	4.56	4.38	*3.63*	4.21	3.94	3.68	3.54	5.47	**3.1**
#4	PSNR↑	12.04	21.28	21.35	23.69	22.02	22.28	23.26	*23.55*	20.23	**25.1**
	SSIM↑	0.5554	0.7589	0.755	*0.828*	0.7711	0.776	0.8142	0.8324	0.719	**0.8579**
	ERGAS↓	122.45	32.86	33.17	22.73	27.66	27.95	24.03	*23.9*	32.53	**19.79**
	SAM↓	32.49	4.76	4.51	3.83	4.84	4.2	4.12	*3.87*	6.92	**3.11**
#5	PSNR↑	9.9	20.15	20.23	*20.95*	19.99	20.09	20.68	22.02	19.21	**22.38**
	SSIM↑	0.5105	0.7167	0.7085	0.7566	0.7379	0.7072	*0.7613*	0.7977	0.7289	**0.8129**
	ERGAS↓	106.94	22.59	22.61	*19.75*	22.42	22.26	20.52	17.5	23.81	**16.96**
	SAM↓	33.68	3.83	*3.75*	3.91	4.25	3.96	3.89	3.42	5.93	**2.95**
#6	PSNR↑	10.29	21.37	21.67	*26.62*	22.3	23.15	23.04	27.86	23.64	**29.44**
	SSIM↑	0.5104	0.7634	0.7525	*0.8993*	0.803	0.7986	0.8239	0.9305	0.8498	**0.94**
	ERGAS↓	109.23	20.33	20.01	*10.68*	18.22	16.24	16.72	9.56	15.11	**7.84**
	SAM↓	33.75	4.16	3.87	*2.39*	4.31	3.25	4.08	2.16	3.92	**1.66**
#7	PSNR↑	11.85	22.77	23.09	*27.13*	24.95	23.89	26.37	28.94	24.97	**29.85**
	SSIM↑	0.6104	0.8232	0.8188	*0.9181*	0.8726	0.8407	0.8999	0.9489	0.8879	**0.9496**
	ERGAS↓	85.62	18.9	18.57	*10.68*	14.29	16.22	12.02	8.98	14	**8.04**
	SAM↓	25.35	3.15	2.92	*2.11*	2.8	2.66	2.46	1.9	3.37	**1.49**
#8	PSNR↑	10.29	24.37	24.56	25.33	24.49	24.49	25.08	*25.25*	20.13	**25.71**
	SSIM↑	0.55	0.8131	0.8092	0.8317	0.8256	0.8058	0.8443	*0.8376*	0.7177	**0.8471**
	ERGAS↓	79.61	11.45	11.33	*10.32*	11.31	11.23	10.45	10.3	18.57	**10.03**
	SAM↓	26.62	1.93	*1.81*	1.67	2.05	1.92	1.88	2.11	4.58	**1.62**
#9	PSNR↑	11.02	22.31	22.42	22.77	22.3	22.27	22.85	*22.85*	18.87	**23.2**
	SSIM↑	0.5809	0.7929	0.7873	0.819	0.809	0.7861	0.8249	*0.82*	0.7283	**0.8385**
	ERGAS↓	82.07	16.9	16.88	15.92	16.75	16.87	*15.61*	15.6	24.02	**15.42**
	SAM↓	26.51	2.72	2.65	2.71	2.86	2.75	*2.69*	3.07	6.24	**2.49**
Mean	PSNR↑	10.79	21.52	21.7	*24.37*	22.32	22.42	23.24	25.17	21.49	**26.14**
	SSIM↑	0.5452	0.7666	0.76	*0.8471*	0.8001	0.7824	0.8275	0.8704	0.7858	**0.883**
	ERGAS↓	103.3	22.34	22.27	*15.36*	19.57	19.45	17.51	14.53	21.11	**12.99**
	SAM↓	30.92	3.83	3.65	*2.86*	3.84	3.36	3.45	2.79	5.03	**2.27**

algorithms SNN, SNNTV, TNN, and TNNTV are less effective in eliminating deadlines. STDTV outperforms SPCTV, S2NTNN, and tCTV in terms of texture recovery.

Table III presents the quantitative evaluation results of various completion methods for the SCL-off problem. The best results are highlighted in bold. It can be observed from the table that the proposed STDTV achieves the best performance. It is important to note that the SCL-off problem, characterized by dense and wide deadlines, is significantly more challenging than the HSI random missing problem. For PSNR, which is averaged over nine time nodes, S2NTNN is the current best method with a value of 25.17 dB, while STDTV achieves a PSNR of 26.14 dB, representing an improvement of 0.97 dB.

B. Real Data Experiments

1) Sentinel-2 Time-Series Cloud Removal: The second dataset was a time-series Sentinel-2A/B L1C images over Mechelen, Belgium. This dataset was provided by Zhang et al. [47]. There are three bands (B5, B7, and B11), and seven time nodes (2018-08-16, 2018-08-31, 2018-09-05, 2018-09-15, 2018-09-25, 2018-10-15, and 2018-10-20), resulting in an image of a size $512 \times 512 \times 3 \times 7$.

Fig. 6 shows the first and the fifth time nodes, before and after reconstruction using different methods. It is shown that SNN and SNNTV are blurred, TNN and tCTV suffer from color distortion, and SPCTV and FCTN lack details. In the opposite, STDTV recoveries more details and textures.

2) SLC-Off Landsat-7 ETM+ Image: The first was a time-series Landsat-7 ETM+ image captured over Zhongwei Ningxia, China. There are six bands (blue, green, red, near-infrared, shortwave infrared 1, and shortwave infrared 2), and 11 time nodes (2004-02-06, 2004-08-16, 2004-10-03, 2004-11-04, 2004-11-20, 2005-04-13, 2005-11-07, 2006-03-15,2006-10-25, 2007-02-14, and 2008-09-12), resulting in an image of a size $512 \times 512 \times 6 \times 11$.

Fig. 7 shows the first time node (2004-02-06) with composited bands 3, 2, and 1, before and after reconstruction using

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

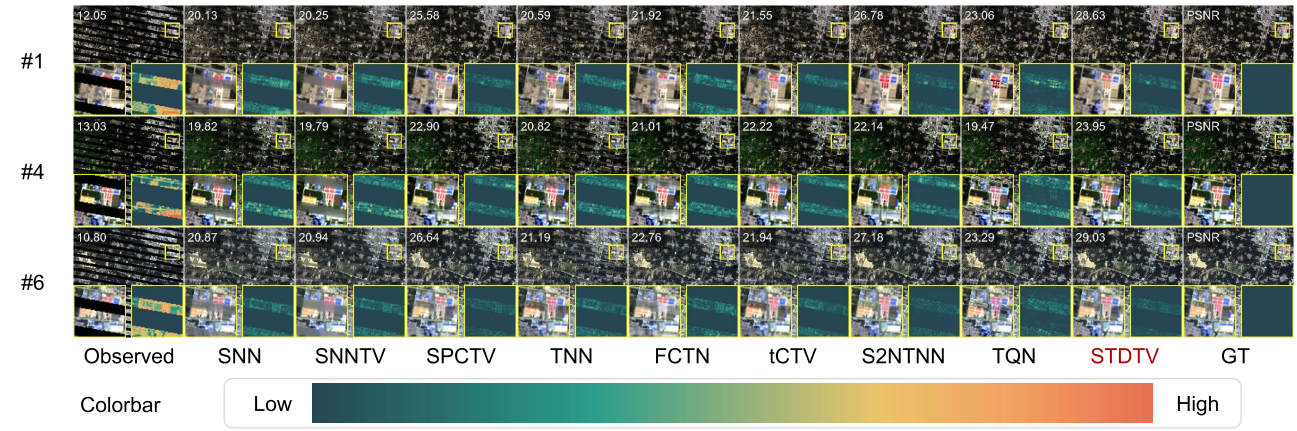


Fig. 5. MTI SLC-off restoration results of different methods (bands: 4-3-2) on the Puyang dataset. (From top to bottom) There are three time-nodes. (Top left) PSNR value of these bands. (Bottom left) and (bottom right) Amplified area and its error map.

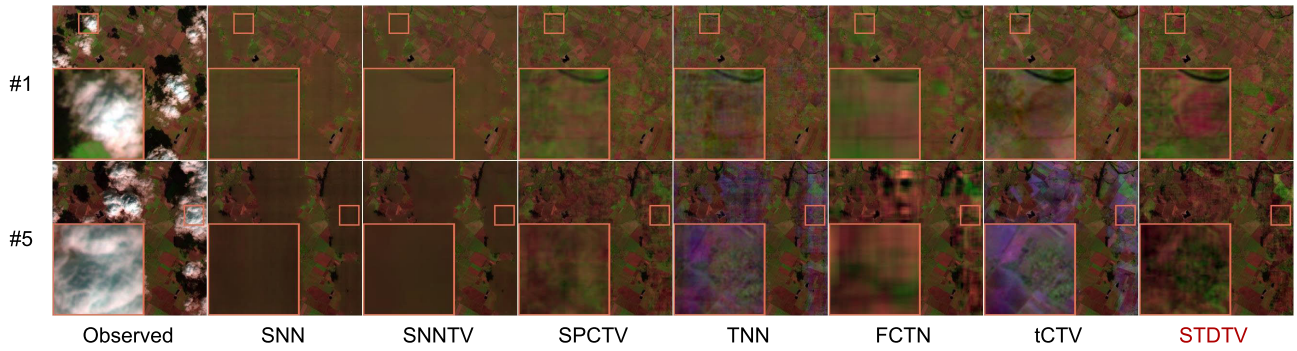


Fig. 6. Recovered effects for cloud removal on the first and the fifth time nodes of Mechelen dataset.

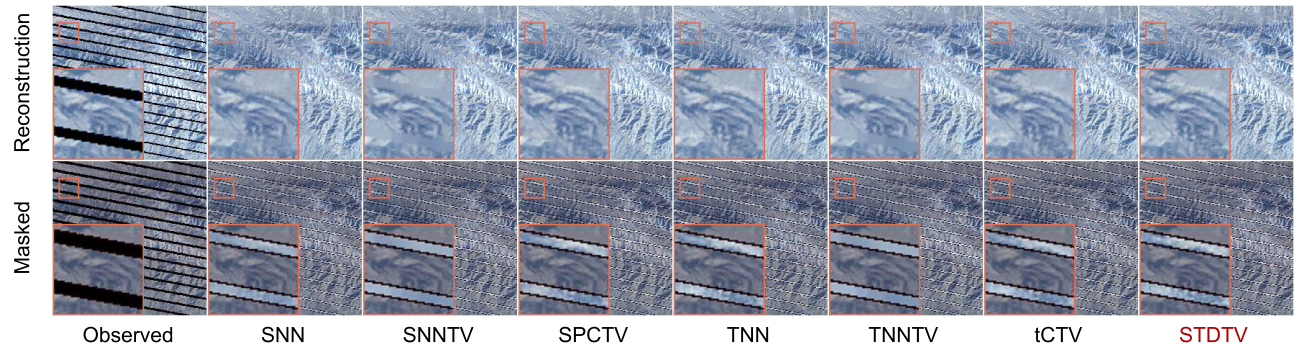


Fig. 7. Recovered effects for SLC-off Landsat-7 ETM+ images on the first time node (2004-02-06) of Zhongwei dataset. (Top) Reconstructed images. (Bottom) Reconstructed images with masks. The areas with gray shadows are observed pixels, and other areas without shadows correspond to missing pixels.

different methods. For better inspection, the reconstructed pixels are highlighted by applying dark masks to observed pixels, as illustrated in Fig. 7 (bottom). It is shown that STDTV recoveries more details and textures.

C. Discussion

1) Sensitivity to Rank: As stated in Section IV-D, setting STD-rank is laborious. With the linear sampling strategy, it only needs to set the rank of the first layer, $(R_{0,1}, R_{0,2}, R_{0,3})$. We assume the rank is proportional to the data size (H, W, B), and the spatial modes (i.e., height and width) have the same proportion coefficient, so we set $(R_{0,1}, R_{0,2}, R_{0,3}) = (s_1H, s_1W, s_2B)$, where s_1 and s_2 are two spatial and spectral proportion coefficients, respectively.

To explore the sensitivity of STDTV to the rank, s_1 and s_2 are set from 0.1 to 1 with a step of 0.1. Fig. 8 shows the PSNR values for each pair of (s_1, s_2). It is shown that different tasks have diverse behaviors. For HSI inpainting, STDTV favors smaller s_1 and s_2, and the recommended values are $s_1 = 0.6$ and $s_2 = 0.1$. For SLC-off inpainting, the PSNR values of STDTV always belong to the interval $[25, 25.6]$, is less sensitive to s_1 and s_2. However, the PSNR values also tend to slightly decrease with s_1 and s_2 growing. The recommended values thus are set to $s_1 = 0.2$ and $s_2 = 0.1$. For cloud removal, STDTV favors the middle s_1 and s_2. The recommended values are $s_1 = 0.4$ and $s_2 = 0.3$.

2) Sensitivity to L: The number of layers L in a STD model is another important parameter. Intuitively, deeper the STD is,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

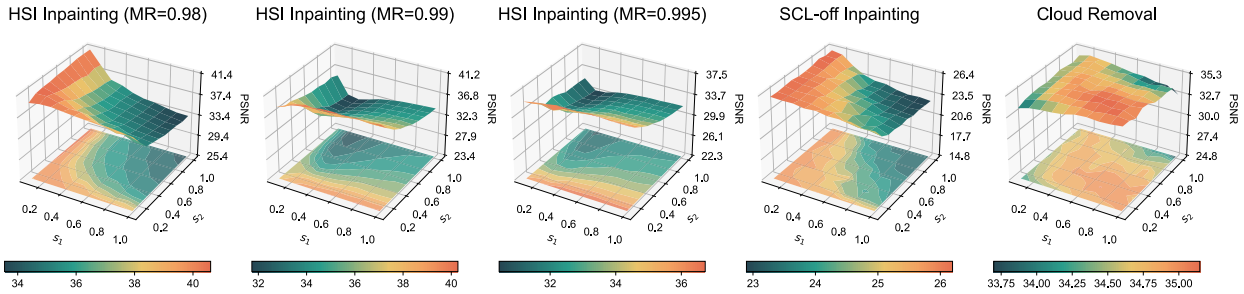


Fig. 8. PSNR surface with difference STD-ranks for difference synthetic experiments.

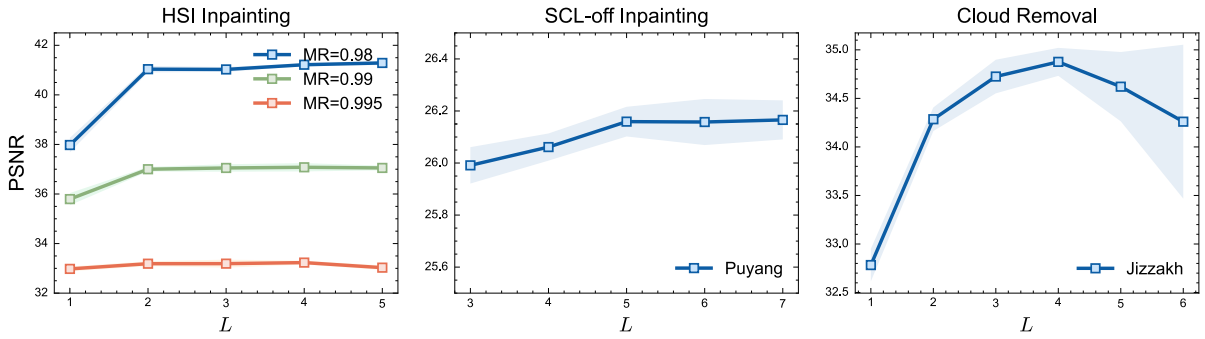


Fig. 9. PSNR curves with difference layer numbers for difference synthetic experiments. The shaded areas are error bars calculated over ten trials.

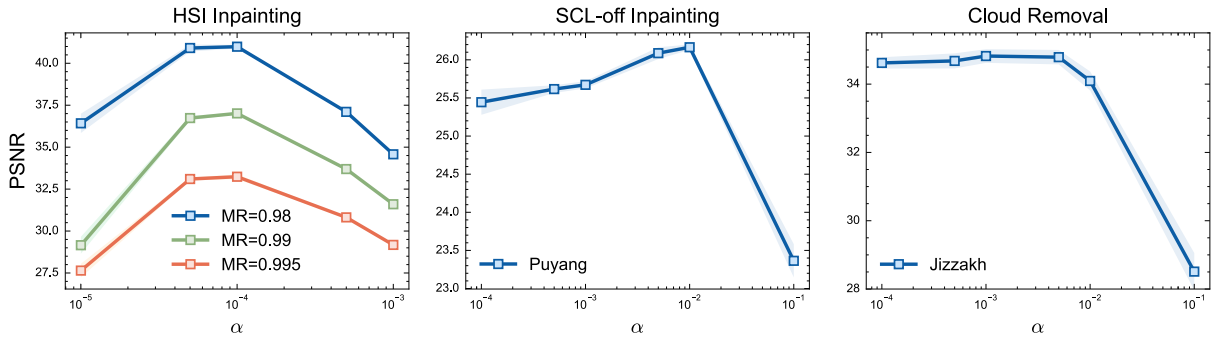


Fig. 10. PSNR curves with varying trade-off parameters of 3DWTV regularizer α for difference synthetic experiments. The shaded areas are error bars calculated over ten trials.

better the performance is. To validate this presumption, the experiments were conducted with varying L. As illustrated in Fig. 9, different tasks have different sensitivities to L. Additionally, there also shows a tendency that larger L leads to a larger volatility, which may be because it is more difficult to train a model with more parameters. L should not be set as a large value.

The layer number, L, is recommended to set as 2, 4, and 5 for HSI inpainting, cloud removal, and SLC-off inpainting, respectively. Although L depends on the type of tasks, a conclusion can be drawn that the optimal value for L tends to smaller when the degradation is more significant. The reason may be that the severe degradation hinders STD from obtaining useful information to recover missing pixels, and thus STD has to increase the number of layers to obtain more powerful capability to learn knowledge from complicated data.

3) Sensitivity to α: Subsequently, the influence of TV regularizer's weight α is studied. As shown in Fig. 10, TV regularizer prompts better performances. α is recommended to set as 10^{-4}, 10^{-2}, and 10^{-3} for HSI inpainting, SLC-off inpainting, and cloud removal, respectively.

TABLE IV
PARAMETER CONFIGURATION FOR THREE TASKS

	HSI Inpainting	Cloud Removal	SCL-off Inpainting
α	0.0001	0.001	0.01
β_3	1	0.05	0.05
L	2	4	5
(s_1, s_2)	(0.6,0.1)	(0.4,0.3)	(0.2,0.1)

4) Sensitivity to β_3: In 3DWTV, β_1 and β_2 correspond to spatial smoothness and are often set as 1. However, β_3 controls the weight for spectral smoothness and needs tuning. As shown in Fig. 11, β_3 depends on the data type. HSIs have stronger smoothness along the spectral dimension, so $\beta_3 = 1$ leads to the best result. In opposite, SCL-off inpainting and cloud removal tasks operate on MTIs with weaker smoothness along channel dimension, so $\beta = 0.05$ is a better choice. As a summary, Table IV lists the recommended parameters for the three tasks.

5) Ablation Studies: The ablation experiments focus on the influence of nonlinear function $\sigma(\cdot)$ and 3DWTV.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

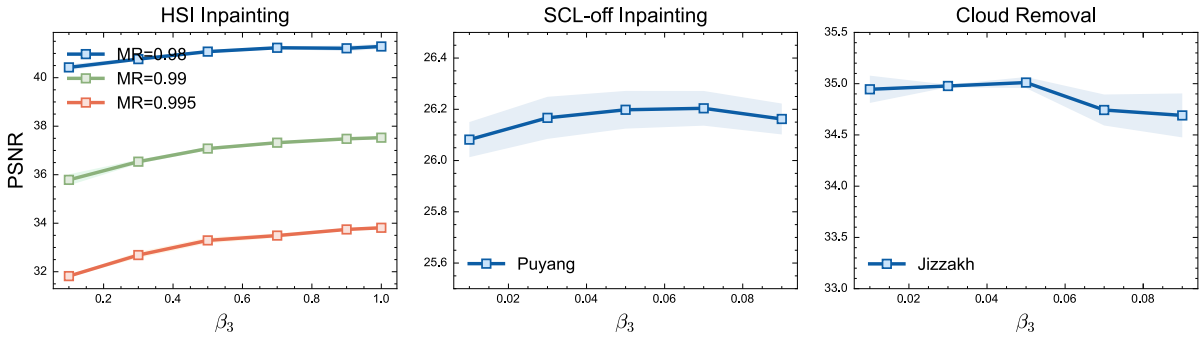


Fig. 11. PSNR curves with varying weights of 3DWTV regularizer β for difference synthetic experiments. The shaded areas are error bars calculated over ten trials.

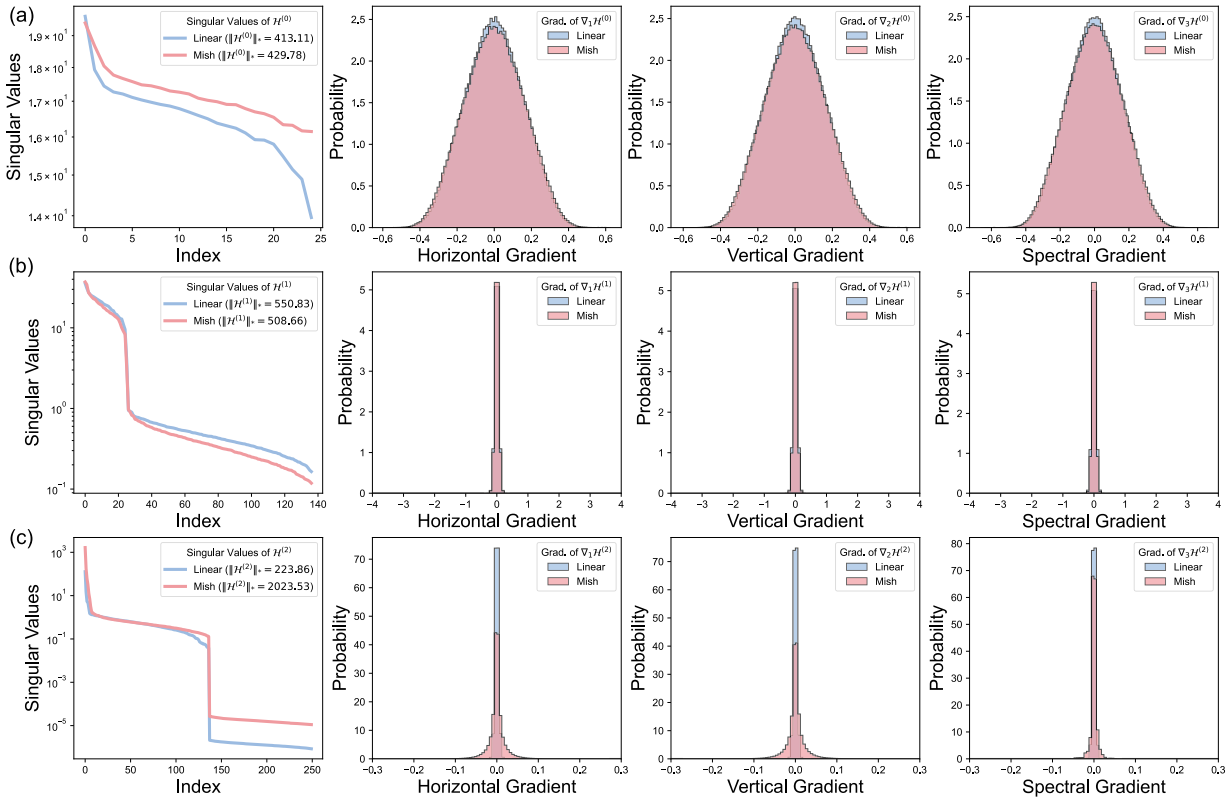


Fig. 12. Low-rankness (the first column) and smoothness of (other columns) (a) $\mathcal{H}^{(0)}$, (b) $\mathcal{H}^{(1)}$, and (c) $\mathcal{H}^{(2)}$.

TABLE V
PSNR PERFORMANCE OF STDTV WITH LINEAR AND NONLINEAR FUNCTIONS ON HSI INPAINTING TASKS

Cases	$\sigma(\cdot)$	3DWTV	98%	99%	99.5%
I	Linear	$\surd$	39.99	37.07	33.62
II	Mish	$\times$	36.95	21.68	17.52
Ours	Mish	$\surd$	40.77	37.65	34.09

Case I: As shown in Case I of Table V, performances of STDTV with the linear function and Mish implies that the nonlinear function indeed booms the performance To further investigate how nonlinear functions change the STDTV model, Fig. 12 displays the singular value curves and gradient map histograms of $\mathcal{H}^{(i)}(i = 0, 1, 2)$ for the HSI inpainting task with MR=0.98. It is shown that the nuclear norm of $\mathcal{H}^{(i)}$ for linear functions and Mish has similar values. However, the singular value curve of $\mathcal{H}^{(2)}$ displays that Mish gives greater values in the tail, indicating that more details are preserved by Mish. From the gradient map histograms, the similar phenomena is observed that the linear function and Mish share the similar smoothness for $\mathcal{H}^{(0)}$ and $\mathcal{H}^{(1)}$. However, the gradient map of $\mathcal{H}^{(2)}$ for the linear function is extremely sparser than Mish, so the linear function may lose important textures.

Case II: STDTV would degrade as the STD model if 3DWTV is not integrated. There is no doubt that STDTV significantly improves the performance, especially for high missing rates. Please recall Fig. 10 which studies the sensitivity of 3DWTV's regularization parameter. Even with a very small value (e.g., $\alpha = 10^{-5}$), STDTV still outperforms STD, proving the importance of 3DWTV. The reason may be that missing rate cases lacks enough supervision information for STD which contains numerous parameters.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

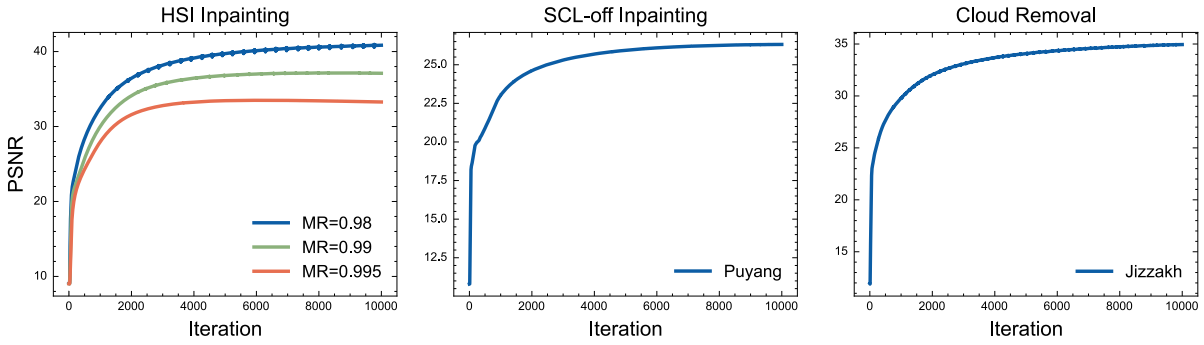


Fig. 13. PSNR curves versus iterations.

TABLE VI
EXECUTION TIME (IN SECONDS) OF DIFFERENT METHODS ON HSI INPAINTING TASKS

Methods	SNNTV	SPCTV	FCTN	tCTV	TQN	STDTV
Time	1690	4478	438	1596	332	2466

6) Execution Time: Table VI reports the execution time for representative methods. It is shown that the methods with better performances tend to be more time-consuming. STDTV cannot be recognized as a very fast method, due to numerous parameters within it.

7) Convergence: We adopted experiments to demonstrate the convergence of STDTV. Fig. 13 shows the changes in PSNR values for STDTV, with the increase in iteration number under different simulated experiments. As the iteration number increases, the PSNR increases to stable values in all cases, indicating convergence of STDTV.

8) Limitations and Future Work: Based on the above analysis, STDTV, despite its high performance, is not efficient in terms of execution time and optimization. Accelerating the optimization algorithm or exploring lightweight variant for STDTV remains the important future work.

The number of layers, L, is another important parameter. It is demonstrated that different tasks may favor different choice for L, and thus it is still a challenge to determine optimal choice of L, which remains a future work.

VI. CONCLUSION

In this study, a NTD model is developed by proposing a multi-nonlinear product to replace the multilinear product (a.k.a., mode-n product). Inspired by the human visual system, the NTD, serving as the fundamental computation unit, is extended as a STD model. For RS imagery inpainting problems, the STD combined with total variation regularizer is formulated, and it exhibits superior performance to SOTA methods for HSI random inpainting, SLC-off inpainting, and MTI cloud removal tasks.

REFERENCES

[1] J. Wang, Z. Cui, T. Jiang, C. Cao, and Z. Cao, "Lightweight deep neural networks for ship target detection in SAR imagery," *IEEE Trans. Image Process.*, vol. 32, pp. 565–579, 2023.

[2] Q. Wang, W. Huang, Z. Xiong, and X. Li, "Looking closer at the scene: Multiscale representation learning for remote sensing image scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 4, pp. 1414–1428, Apr. 2022.

[3] S. Xu, X. Cao, J. Peng, Q. Ke, C. Ma, and D. Meng, "Hyperspectral image denoising by asymmetric noise modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5545214.

[4] S. Xu, J. Zhang, and C. Zhang, "Hyperspectral image denoising by low-rank models with hyper-laplacian total variation prior," *Signal Process.*, vol. 201, Dec. 2022, Art. no. 108733.

[5] Y.-B. Zheng, T.-Z. Huang, X.-L. Zhao, Q. Zhao, and T.-X. Jiang, "Fully-connected tensor network decomposition and its application to higher-order tensor completion," in *Proc. Conf. AAAI Artif. Intell.*, May 2021, vol. 35, no. 12, pp. 11071–11078.

[6] T. Ji, N. Yokoya, X. X. Zhu, and T. Huang, "Nonlocal tensor completion for multitemporal remotely sensed images' inpainting," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3047–3061, Jun. 2018.

[7] S. Xu, J. Zhang, Z. Zhao, K. Sun, J. Liu, and C. Zhang, "Deep gradient projection networks for pan-sharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1366–1375.

[8] S. Xu, O. Amira, J. Liu, C.-X. Zhang, J. Zhang, and G. Li, "HAM-MFN: Hyperspectral and multispectral image multiscale fusion network with RAP loss," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4618–4628, Jul. 2020.

[9] C.-H. Lin, K.-H. Lai, Z.-B. Chen, and J.-Y. Chen, "Patch-based information reconstruction of cloud-contaminated multitemporal images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 163–174, Jan. 2014.

[10] L. Wang, J. J. Qu, X. Xiong, X. Hao, Y. Xie, and N. Che, "A new method for retrieving band 6 of Aqua MODIS," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 2, pp. 267–270, Apr. 2006.

[11] C. Zeng, H. Shen, and L. Zhang, "Recovering missing pixels for Landsat ETM+ SLC-off imagery using multi-temporal regression analysis and a regularization method," *Remote Sens. Environ.*, vol. 131, pp. 182–194, Apr. 2013.

[12] A. Cichocki, N. Lee, I. Oseledets, A. H. Phan, Q. Zhao, and D. P. Mandic, "Tensor networks for dimensionality reduction and large-scale optimization: Part 1 low-rank tensor decompositions," *Found. Trends Mach. Learn.*, vol. 9, nos. 4–5, pp. 249–429, 2016.

[13] W.-J. Zheng, X.-L. Zhao, Y.-B. Zheng, and Z.-F. Pang, "Nonlocal patch-based fully connected tensor network decomposition for multispectral image inpainting," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[14] W.-J. Zheng, X.-L. Zhao, Y.-B. Zheng, J. Lin, L. Zhuang, and T.-Z. Huang, "Spatial–spectral–temporal connective tensor network decomposition for thick cloud removal," *ISPRS J. Photogramm. Remote Sens.*, vol. 199, pp. 182–194, May 2023.

[15] Z.-C. Wu, T.-Z. Huang, L.-J. Deng, H.-X. Dou, and D. Meng, "Tensor wheel decomposition and its tensor completion application," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 27008–27020.

[16] C. Yu, L. Chen, L. Su, M. Fan, and S. Li, "Kriging interpolation method and its application in retrieval of MODIS aerosol optical depth," in *Proc. 19th Int. Conf. Geoinformat.*, Jun. 2011, pp. 1–6.

[17] Q. Cheng, H. Shen, L. Zhang, and P. Li, "Inpainting for remotely sensed images with a multichannel nonlocal total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 175–187, Jan. 2014.

[18] A. Maalouf, P. Carré, B. Augereau, and C. Fernandez-Maloigne, "A bandelet-based inpainting technique for clouds removal from remotely sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 7, pp. 2363–2371, Jul. 2009.

[19] A. Criminisi, P. Perez, and K. Toyama, "Region filling and object removal by exemplar-based image inpainting," *IEEE Trans. Image Process.*, vol. 13, no. 9, pp. 1200–1212, Sep. 2004.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.

[20] P. Rakwatin, W. Takeuchi, and Y. Yasuoka, "Restoration of Aqua MODIS band 6 using histogram matching and local least squares fitting," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 2, pp. 613–627, Feb. 2009.

[21] H. Shen, C. Zeng, and L. Zhang, "Recovering reflectance of Aqua MODIS band 6 based on within-class local fitting," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 4, no. 1, pp. 185–192, Mar. 2011.

[22] I. Gladkova, M. D. Grossberg, F. Shahriar, G. Bonev, and P. Romanov, "Quantitative restoration for MODIS band 6 on Aqua," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 6, pp. 2409–2416, Jun. 2012.

[23] X. Li, H. Shen, L. Zhang, H. Zhang, and Q. Yuan, "Dead pixel completion of Aqua MODIS band 6 using a robust M-estimator multiregression," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 4, pp. 768–772, Apr. 2014.

[24] J. Wang, P. A. Olsen, A. R. Conn, and A. C. Lozano, "Removing clouds and recovering ground observations in satellite image sequences via temporally contiguous robust matrix completion," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2754–2763.

[25] J. Lin, T.-Z. Huang, X.-L. Zhao, Y. Chen, Q. Zhang, and Q. Yuan, "Robust thick cloud removal for multitemporal remote sensing images using coupled tensor factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5406916.

[26] T. Jiang, X. Zhao, H. Zhang, and M. K. Ng, "Dictionary learning with low-rank coding coefficients for tensor completion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 2, pp. 932–946, Feb. 2023.

[27] Q. Zhao, G. Zhou, L. Zhang, A. Cichocki, and S.-I. Amari, "Bayesian robust tensor factorization for incomplete multiway data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 4, pp. 736–748, Apr. 2016.

[28] T. Yokota, Q. Zhao, and A. Cichocki, "Smooth PARAFAC decomposition for tensor completion," *IEEE Trans. Signal Process.*, vol. 64, no. 20, pp. 5423–5436, Oct. 2016.

[29] M. K. Ng, Q. Yuan, L. Yan, and J. Sun, "An adaptive weighted tensor completion method for the recovery of remote sensing images with missing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3367–3381, Jun. 2017.

[30] T.-H. Zhang, J.-L. Zhao, S. Fang, Z. Li, and M.-G. Gong, "Full-Mode-Augmentation tensor-train rank minimization for hyperspectral image inpainting," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5500913.

[31] W. He, N. Yokoya, L. Yuan, and Q. Zhao, "Remote sensing image reconstruction using tensor ring completion and total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8998–9009, Nov. 2019.

[32] Q. Zhao, G. Zhou, S. Xie, L. Zhang, and A. Cichocki, "Tensor ring decomposition," *CoRR*, vol. abs/1606.05535, pp. 1–14, Jun. 2016.

[33] Y. Qiu, G. Zhou, Q. Zhao, and S. Xie, "Noisy tensor completion via low-rank tensor ring," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1127–1141, Jan. 2024.

[34] Z. Long, C. Zhu, J. Liu, and Y. Liu, "Bayesian low rank tensor ring for image recovery," *IEEE Trans. Image Process.*, vol. 30, pp. 3568–3580, 2021.

[35] Y. Chen, J. Zeng, W. He, X.-L. Zhao, and T.-Z. Huang, "Hyperspectral and multispectral image fusion using factor smoothed tensor ring decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515417.

[36] H. Xu, M. Qin, Y. Yan, M. Zhang, and J. Zheng, "Nonlocal B-spline representation of tensor decomposition for hyperspectral image inpainting," *Signal Process.*, vol. 206, May 2023, Art. no. 108888.

[37] D. Misra, "Mish: A self regularized non-monotonic activation function," in *Proc. 31st Brit. Mach. Vis. Conf.* Durham, U.K.: BMVA Press, Sep. 2020, pp. 1–14.

[38] M. Riesenhuber and T. Poggio, "Hierarchical models of object recognition in cortex," *Nature Neurosci.*, vol. 2, no. 11, pp. 1019–1025, Nov. 1999.

[39] G. Trigeorgis, K. Bousmalis, S. Zafeiriou, and B. W. Schuller, "A deep matrix factorization method for learning attribute representations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 3, pp. 417–429, Mar. 2017.

[40] J. Liu, P. Musialski, P. Wonka, and J. Ye, "Tensor completion for estimating missing values in visual data," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 208–220, Jan. 2013.

[41] X. Li, Y. Ye, and X. Xu, "Low-rank tensor completion with total variation for visual data inpainting," in *Proc. 31st AAAI Conf. Artif. Intell. (AAAI)*, 2017, pp. 2210–2216.

[42] C. Lu, J. Feng, Z. Lin, and S. Yan, "Exact low tubal rank tensor recovery from Gaussian measurements," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, J. Lang, Ed., Jul. 2018, pp. 2504–2510.

[43] H. Wang, J. Peng, W. Qin, J. Wang, and D. Meng, "Guaranteed tensor recovery fused low-rankness and smoothness," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10990–11007, Sep. 2023.

[44] Y. Luo, X. Zhao, T. Jiang, Y. Chang, M. K. Ng, and C. Li, "Self-supervised nonlinear transform-based tensor nuclear norm for multi-dimensional image recovery," *IEEE Trans. Image Process.*, vol. 31, pp. 3793–3808, 2022.

[45] H. Kong, C. Lu, and Z. Lin, "Tensor Q-rank: New data dependent definition of tensor rank," *Mach. Learn.*, vol. 110, no. 7, pp. 1867–1900, Jul. 2021.

[46] Y. Cen et al., "Aerial hyperspectral remote sensing classification dataset of Xiongan New Area (Matiwan Village)," *J. Remote Sens.*, vol. 24, no. 11, pp. 1299–1306, 2020.

[47] Q. Zhang, Q. Yuan, J. Li, Z. Li, H. Shen, and L. Zhang, "Thick cloud and cloud shadow removal in multitemporal imagery using progressively spatio-temporal patch group deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 148–160, Apr. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 08:43:25 UTC from IEEE Xplore. Restrictions apply.